RAI Risk Taxonomy 2.0: Top-down evidence selection and staged classification of 1,711 global AI risks

RAI Risk Taxonomy Project

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Abstract

We present a reproducible, top-down account of the evidence and classification pipeline used to construct RAI Risk Taxonomy 2.0. The integrated evidence space contains 3,906,767 AI records: 1,749,159 papers, 125,372 policy reports, and 2,032,236 patents. A curated AI-risk evidence view contains 82,971 records (2.1% of the AI evidence space). Canonicalization, source crosswalking, and risk-card construction yields 1,726 candidate rows. A conservative exact-match check merges one duplicate, yielding 1,725 candidate L4 cards. A taxonomy-level check against the 24 authoritative Physical AI L3 labels excludes 14 exact label leaks, yielding 1,711 L4 cards. A staged hierarchy-blind classification procedure initially assigns all cards to 50 L3 nodes. Subsequent definition and mapping audits, followed by a conservative v2.8.0 Agentic expansion, temporarily produced 54 L3 nodes and preserved an operational path for every card. Release v2.14.0 retires the four sparse experimental Agentic families, archives their identities and definitions, and force-migrates their 39 cards to the best available active families. It contains 1,193 General, 336 Agentic, and 182 locked Physical cards across 50 semantic L3 nodes. Release v2.15.0 then adds domain-specific HOLD review paths under General and Agentic AI, moves 626 and 94 review cards respectively, and preserves each original semantic L3 as lineage metadata. Incremental constrained-EM audits preserve the authoritative Physical set and move only definition- and mechanism-supported boundary cases. A subsequent Memory-card scope audit preserves all identities and paths while differentiating three overlapping L4 intensions. All forced migrations remain under review; 720 cards (42.1%) are marked `HOLD` for human taxonomic review. A full-release BGE-M3 rerun finds statistically non-random semantic structure and strong local support for the taxonomy, but does not establish global reliability of the consolidated 50-family assignment when HOLD remains embedded (64.7% top-1 containment; 72.8% perturbation agreement at $\sigma = 0.05$). Re-evaluation of the 991 non-HOLD cards after separation yields 75.8% top-1 containment and 78.4% agreement, a conditional subset result rather than a like-for-like accuracy gain. We specify every evidence filter, classification rule, and validation limit.

Keywords: AI risk taxonomy; evidence synthesis; open-set classification; semantic mapping; Physical AI; responsible AI.

1 Introduction

A taxonomy can appear complete while concealing uncertainty in its source corpus, unit of analysis, or placement rules. RAI Risk Taxonomy 2.0 therefore separates four quantities that must not be conflated: the size of the integrated AI evidence space, the size of the AI-risk evidence view, the number of canonical L4 risk cards, and the number of L4 cards assigned to L3. The central result is a top-down evidence-to-taxonomy chain of 3,906,767 records, 82,971 AI-risk records, 1,726 candidate L4 rows, 1,725 exact-deduplicated candidates, 1,711 level-validated L4 cards, and 1,711 L3-assigned cards.

The transitions do not all preserve the same statistical unit. The first two stages count documents or patent records. The third and fourth stages count canonical risk concepts. Consequently, the 82,971 evidence records are not a one-to-one parent table from which every L4 card was mechanically generated. They form an auditable evidence view supporting corpus coverage and risk interpretation, whereas the L4 registry integrates risk taxonomies, benchmarks, standards, and linked evidence through a source crosswalk.

2 Top-down dataset construction

2.1 Stage A: integrated AI evidence space

The top-level dataset contains 3,906,767 records from three source families: 1,749,159 papers, 125,372 policy reports, and 2,032,236 patents. Paper records originate from the local global Web of Science AI collections. Policy records combine the reviewed curated AI-policy master with the broader policy-materials collection. Patent records originate from the stage-2 AI patent master.

Records are standardized to a common schema and retained only when a non-empty title is available. Deduplication is performed within document type and analytical category. The ordered key hierarchy is patent application ID for patents, followed by DOI, source record ID, URL, and a normalized title–year–institution fallback. Cross-category overlap is intentionally retained because AI, AI infrastructure, Physical AI, and AI risk are analytical views rather than mutually exclusive partitions.

2.2 Stage B: integrated AI-risk evidence view

The AI-risk evidence view contains 82,971 records: 50,375 papers, 938 policy reports, and 31,658 patents. Its inclusion rule is source-governed rather than a single keyword query. Papers combine curated academic risk references, the clean and integrated AI-risk paper corpora, Physical AI safety benchmarks, and verified paper records from the top-200 evidence set. Policy reports combine curated risk-policy references with verified report records. Patents combine curated risk-patent references with verified patent records. Record-type filters are applied before the same title-validity and within-category deduplication rules used at Stage A.

This view represents 2.1% of the integrated AI evidence space. Because the source collections and analytical categories can overlap, this percentage is a descriptive ratio, not the estimated prevalence of risk research in all AI scholarship.

2.3 Stage C: canonical L4 registry

The unit changes from evidence record to risk concept. Source risk IDs are joined through the source crosswalk, semantically redundant cards are reviewed, and one canonical row is retained per L4 failure mode or harm mechanism. Permanent identifiers `RAI4-0001` through `RAI4-1726` are allocated independently of the risk label. The initial result is 1,726 candidate L4 rows. Automatic deduplication is restricted to pairs whose normalized label and mechanism-only definition are both exactly equal; semantic-similarity thresholds are not used. This merges `RAI4-1606` into `RAI4-1083`, retains both source references, and yields 1,725 exact-deduplicated candidates. The retired ID remains in the crosswalk. The registry includes the 182-card Physical AI taxonomy as a protected mapping set.

The ratio of L4 cards to AI-risk evidence records is 2.1%. It is reported only as a scale-change diagnostic: one document may support multiple risks, and one risk may be supported by multiple documents. It is not a document acceptance rate.

2.4 Stage D: taxonomy-level validation

The deployed Physical AI site is the authoritative source for its 24 L3 names and 182 L4 cards. Its live HTML was byte-identical to the local repository snapshot used for this check. We compare the English component of each authoritative Physical L3 name with every candidate L4 label after NFKC normalization, case folding, ampersand normalization, and removal of non-alphanumeric separators. A card is excluded only when these normalized labels are exactly equal; semantic similarity and definition similarity are not used. Fourteen L4 rows match category names and are retired as taxonomy-level leakage, including `RAI4-0553` “Accidental harm”, which maps to Physical L3 `P3.1`. Their definitions and references remain in the audit crosswalk. This yields 1,711 canonical L4 cards without changing any of the protected 182 Physical L4 mappings.

2.5 Authoritative Physical L4 content synchronization

Release v2.4 synchronizes the content of all 182 protected Physical cards from the local repository underlying the deployed Physical AI Risk Taxonomy site. The immutable RAI4 crosswalk and L3 paths are retained, while the bilingual label and definition, severity, probability, 3H1R attributes, evidence justifications, and 360 reference rows are replaced by the current Physical source values. Of these reference rows, 359 contain a justification and one source row retains an empty justification without imputation. Bilingual terminal parenthetical text is split into Korean and English fields to preserve the global card schema. Impact is recalculated as $I = s\,p$ from the synchronized severity s and probability p. Percentile is not displayed for any card because the cards do not constitute a common measurement population suitable for relative ranking. Existing percentile fields remain provenance data only.

2.6 Complete English–Korean localization

Release v2.5 requires bilingual display at every taxonomy level. The supplied L0–L3 terminology and Korean definitions are used as the controlled glossary. The 182 Physical L4 cards retain their authoritative bilingual source text. For the remaining 1,529 cards, the localization pipeline removes the repeated provenance boilerplate from each English definition, selects the first sentence that states the core failure mechanism, generates a concise Korean definition, and normalizes risk-domain terminology. Context-sensitive corrections include *poisoning* as “data or model contamination” rather than substance addiction, *washing* as false compliance presentation, and *rubber stamping* as nominal approval. Every generated row retains the English source-sentence hash, method, and `human_review_status=pending`. Thus all 1,711 cards contain non-empty English and Korean labels and definitions, while automated Korean drafts remain explicitly reviewable rather than being represented as human-approved ground truth.

2.7 Canonical L2 consolidation

Release v2.6 normalizes L2 to three canonical categories: *Interaction Safety*, *System Safety*, and *Societal Safety*. The former General labels *Interaction* and *System*, and the former Agentic label *System*, are renamed accordingly. Six L1-specific path nodes remain as implementation edges so that the strict L0–L1–L2–L3 tree and all L3 parent paths are preserved; each edge references one of the three canonical L2 identifiers. Thus the reported L2 category count is three, while no L3 path or L4 assignment changes. Release v2.15.0 adds a fourth canonical display category, HOLD, with General and Agentic path nodes. This fourth category is a review-state overlay rather than a semantic safety dimension.

2.8 Stable L4 identifiers and L1 display semantics

The public explorer keeps the stable `RAI4-####` identifier independent of taxonomy placement. L1 membership is communicated through an explicit text badge, a consistent icon, and a visual color: a brain and blue for General-purpose AI, a compass and green for Agentic AI, and a robot and red for Physical AI. The icon–color pair is repeated in the global navigation, filters, hierarchy panel, and L4 cards. The color is derived from the card’s current L1 path at render time rather than encoded into the permanent identifier. This allows later human-approved remapping without identifier churn and avoids using color or icon as the sole carrier of categorical information.

2.9 Stage E: L3 assignment policy and later audits

The operational policy assigns all 1,711 canonical L4 cards to an L3 node (100.0%). The original three-stage classifier used 50 L3 nodes and retained 76 decision-review markers after level validation. Release v2.7.0 then removed 1,129 instances of circular taxonomy prose from non-Physical definitions, performed evidence-constrained mapping review, and marked 692 cards where the available definition and evidence did not justify an approved taxonomic fit. The v2.8.0 Agentic expansion evaluates every non-Physical card while locking all 182 Physical mappings. It moves 31 cards into four new L3 nodes and clears seven review markers, leaving 685 cards (40.0%) marked `HOLD`. Releases v2.9.0–v2.12.0 then audit Anthropomorphism, Goal Misalignment, and Copyrights, leaving 689 cards (40.3%) marked `HOLD`. Release v2.13.0 differentiates three overlapping Memory cards and places all three under source-entailment or near-duplicate scope review, yielding 692 held cards (40.4%). Release v2.14.0 then retires the four sparse Agentic families and force-migrates all 39 affected cards; 28 newly enter review, yielding 720 held cards (42.1%). The marker is internal review metadata on an already assigned card, not a separate L1, L2, or L3 category.

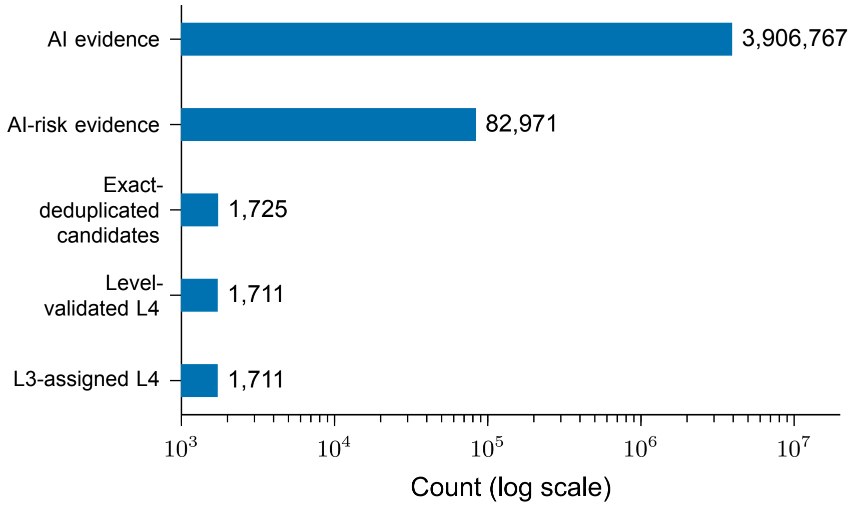

Figure 1: **Top-down evidence and taxonomy funnel.** Counts move from document/patent records to canonical risk concepts at the transition from AI-risk evidence to L4. The logarithmic axis preserves visibility across four orders of magnitude.

Table 1: Top-down stages, counts, and inclusion filters.

Stage	Count	Unit	Principal filter
AI evidence	3,906,767	Document or patent record	AI source collections; valid title; standardized schema; within-type/category deduplication.
AI-risk evidence	82,971	Document or patent record	Curated/integrated risk sources; record-type inclusion; valid title; within-type/category deduplication.
Exact-deduplicated candidates	1,725	Risk card	Exact normalized label and core-definition match only; one retired ID mapped to its canonical card.
Canonical L4	1,711	Risk card	Exact normalized-label exclusion against 24 authoritative Physical L3 names; 14 level leaks retired; all 182 Physical L4 cards preserved.
L3 assigned	1,711	Risk card	Protected Physical mapping or staged non-Physical placement; 689 decision-required cards remain assigned and marked `HOLD`.

3 Mathematical classification procedure

3.1 Protected Physical AI mapping

The 182 L4-to-L3 mappings inherited from the Physical AI taxonomy are treated as the locked gold set. Later stages cannot change these paths. Before the taxonomy-level exclusion, the remaining 1,543 candidates enter the hierarchy-blind classification procedure. Legacy global L0–L3 labels, source families, and source-ID prefixes remain reference metadata and are excluded from prediction features.

3.2 Stage 1: strict open-set gate

The input for each non-Physical card is the L4 label, a mechanism-only definition with hierarchy restatements removed, and the evidence title. Let s_{sem} be the top-1 semantic score, s_{comp} the top-1 composite score, m_{sem} and m_{comp} their respective top-1 margins, and v_{anchor} the number of anchor views selecting the same L3. A candidate is retained only when all seven conditions hold:

1. exactly one L3 is rule-eligible;
2. the rule winner, semantic top-1, and composite top-1 are identical;
3. $s_{sem} \geq 0.55$ and $s_{comp} \geq 0.54$;
4. $m_{sem} \geq 0.035$ and $m_{comp} \geq 0.035$;
5. $v_{anchor} \geq 2$ across three anchor views;
6. no gap sentinel, multi-mechanism flag, or Physical-boundary flag is present; and
7. two independent expert agents approve the same exact L3.

This gate yields 37 new expert-consensus proposals. Together with the 182 Physical locks, Stage 1 assigns 219 cards and withholds 1,507.

3.3 Stage 2: suitability-ranked relaxation

Stage 2 preserves all 219 protected assignments and computes

$$S_2 = 0.55q_{sem} + 0.20q_{sm} + 0.10q_{cm} + 0.10q_{vote} + 0.05q_{rule} - p_{gap}, \tag{1}$$

where

$$\begin{aligned}
q_{sem} &= \text{clip}\left(\frac{s_{sem} - 0.45}{0.25}, 0, 1\right), &(2)\\
q_{sm} &= \text{clip}\left(\frac{m_{sem}}{0.05}, 0, 1\right), &(3)\\
q_{cm} &= \text{clip}\left(\frac{m_{comp}}{0.05}, 0, 1\right), &(4)\\
q_{vote} &= v_{anchor}/3, &(5)\\
q_{rule} &= \mathbf{1}[\text{at least one eligible L3}], &(6)\\
p_{gap} &= 0.05\,\mathbf{1}[\text{gap sentinel present}]. &(7)
\end{aligned}$$

S_2 is a deterministic ranking score rather than a calibrated probability. Stage 2 first preserves 55 hard holds. It then orders the remaining holds by ascending S_2, with permanent RAI4 ID as the tie breaker, and reserves the lowest 118. Thus $55 + 118 = 173$ cards remain unresolved and 1,553 cards are assigned (90.0%).

3.4 Stage 3: forced-path test and final policy

Stage 3 preserves the 1,553 Stage 2 paths and assigns only the 173 unresolved cards to the hierarchy-blind top-1 L3 already stored in the score artifact. This produces 1,726 operational paths but does not validate taxonomy fit. Each new path retains its original hold reason, score, review priority, and `human_approved=false` marker.

The final policy marks five difficult-case families for decision review: 23 Physical-scope reviews, 17 expert rejections, 7 expert disagreements, 6 multi-mechanism cases, and 2 low-absolute-fit cases. Their forced L3 path remains populated and `decision_required=true`. The remaining 118 quota holds also retain their forced paths and review metadata.

3.5 Independent expert card review

Two reviewers independently inspected all 1,725 cards for agreement among the L4 label, mechanism definition, evidence, and assigned L3. Reviewer A proposed 30 amendments and Reviewer B proposed 14. Automatic amendment required both reviewers to propose the same label and L3 at high confidence. One card met this rule: `RAI4-0888` changed from "Privacy concerns" to "Anthropomorphic trust-induced privacy disclosure", while retaining `RAI3-G-INT-10`. Its original label remains in provenance metadata. Forty-two cards had a proposal from only one reviewer or incompatible proposals; their labels and paths were not changed. Thirteen were already marked `HOLD` and 29 newly received the marker. The consensus amendment cleared the prior hold on `RAI4-0888`, giving $55 - 1 + 29 = 83$ final `HOLD` cards. No protected Physical AI card was changed.

The later taxonomy-level validation supersedes the proposed adjudication of `RAI4-0553`: "Accidental harm" is itself the authoritative Physical L3 category `P3.1`, not an L4 failure mode. It is therefore excluded rather than renamed or remapped. Thirteen additional exact Physical-L3 label leaks are handled identically. Six of the excluded rows previously carried `HOLD`, reducing the displayed count from 82 to 76. The protected Physical set remains exactly 182 cards.

3.6 v2.7 definition audit and v2.8 Agentic expansion

The v2.7 audit treats L3 load as a triage signal rather than evidence of semantic fit. It removes 1,129 imported hierarchy-name scaffolds from L4 definitions, applies nine evidence-constrained remaps, and marks 692 cards for decision review. Release v2.8.0 then adds four Agentic-specific L3 families: *Goal & Planning*, *Tool Calling*, *Memory*, and *Oversight & Control*. Each node has an English–Korean definition and explicit references. A non-Physical card moves only when its mechanism is intrinsically Agentic, the new family is more specific than every existing General or Agentic alternative, and the necessity gate rejects adequate coverage by the prior taxonomy. Formally, for card i and new family k,

$$k_i^* = \arg\max_{k \in K_{new}} s(i,k), \qquad \Delta_i = s(i,k_i^*) - \max_{j \in K_{old}} s(i,j), \tag{8}$$

and movement additionally requires mechanism-specific rules, sufficient absolute fit, positive separation, and no Physical lock. The deterministic iteration changes 31 paths on pass 1 and zero on pass 2. Counts in the new families are 11, 11, 3, and 6, respectively. All other placements are preserved, and the v2.8.0 review queue contains 685 cards.

3.7 Incremental constrained-EM audits, v2.9.0–v2.12.0

Later audits do not apply unconstrained nearest-centroid output. Each audit allows only the selected source L3 cards to move, treats all other cards as fixed anchors, excludes every Physical destination, and recomputes candidate centroids until no assignment changes. The composite score is

$$S(i,k) = 0.60\, c(i,k) + 0.30\, d(i,k) + 0.10\, w(i,k), \tag{9}$$

where c is current-centroid cosine similarity, d is L3-definition cosine similarity, and w is TF–IDF keyword cosine similarity. A destination must improve on the source by at least 0.020 and have keyword cosine of at least 0.015. Agentic destinations additionally require an explicit autonomous execution mechanism, and destination-specific definition guards reject matches caused by generic shared terms.

The cumulative audits move three Anthropomorphism cards in v2.9.0, five Goal Misalignment cards in v2.10.0, four Copyrights cards in v2.11.0, and eight additional Anthropomorphism cards in v2.12.0. Every such move remains `HOLD` until human adjudication. The final domain counts are 1,165 General, 364 Agentic, and 182 Physical cards; the review queue contains 689 cards.

3.8 Memory-card scope differentiation, v2.13.0

A near-duplicate audit found that the three cards assigned to Agentic Memory had pairwise BGE-M3 cosines of 0.822–0.874, placing every pair above the 99.98th percentile of all released card pairs. In particular, `RAI4-0011` and `RAI4-1670` expressed a broad–narrow subsumption relationship rather than clearly distinct atomic risks. Release v2.13.0 therefore preserves all three immutable IDs, references, impact metrics, and L3 paths, while revising their intensions as follows.

Table 2: Source-scoped differentiation of the three Agentic Memory cards.

L4 ID	Revised label	Distinguishing mechanism
RAI4-0011	Agent context poisoning	Adversarial corruption of retained or retrievable summaries, embeddings, RAG entries, or shared contextual state, without requiring alteration of a dedicated long-term store
RAI4-0484	Unsafe memory accumulation	False, private, stale, or malicious records accumulate because validation, provenance, retention, or deletion is inadequate; a deliberate poisoning attack is not required
RAI4-1670	Persistent agent-memory poisoning	Adversarial insertion or modification of records in dedicated long-term, cross-session memory, with persistent effects on later retrieval, planning, and action

The revisions are editorial hypotheses subject to source-entailment review, not a declaration that the original source rows were independent risks. All three cards are therefore marked `HOLD`. If reviewers cannot support the stated boundaries directly from the evidence, the conservative alternative is canonical consolidation with aliases and preserved source provenance rather than further lexical renaming.

3.9 Sparse Agentic-family retirement and forced migration, v2.14.0

The four experimental Agentic families introduced in v2.8.0 contained only 39 cards in v2.13.0: Goal & Planning (16), Tool Calling (11), Memory (3), and Oversight & Control (9). Release v2.14.0 removes these nodes from the active hierarchy without erasing their history. Their identifiers, bilingual labels, definitions, references, source counts, and introduction metadata are archived in a machine-readable retirement record; the identifiers are explicitly prohibited from reuse. All L4 identifiers, definitions, references, risk metrics, and provenance are preserved.

Every affected card is force-migrated to the most defensible active non-Physical family under a mechanism-first review. Candidate destinations are restricted to the 26 active non-Physical L3 families. BGE-M3 seed similarity is used as a diagnostic ranking, while explicit mechanism rules distinguish goal failure, unsafe execution authority, network propagation, privacy, excessive capability use, misinformation, overconfidence, and contestability. No Physical path is eligible. Because this operation is a forced operational consolidation rather than human adjudication, all 39 migrations carry `HOLD`; 28 were newly marked and 11 retained an existing marker.

Table 3: Destinations of the 39 v2.14.0 forced migrations.

Active destination L3	Cards
Goal Misalignment	15
Excessive Authority	9
Misinformation/Disinformation	4
Non-Contestability	4
Overconfidence	3
Network Effects	2
Privacy	1
Over-Extension	1

The retired families are not declared conceptually invalid. Their sparse granularity is retained as an auditable design experiment that can be restored only through a later, human-approved taxonomy decision. Release v2.14.0 has 50 active L3 nodes, while the four retired nodes remain available solely as archived lineage metadata.

3.10 Domain-specific HOLD review paths, v2.15.0

Release v2.15.0 makes review status navigable in the hierarchy by introducing one HOLD L2 path under General and one under Agentic AI. Because L4 cards must terminate at L3, each path contains one child node, *Taxonomy Decision Hold*. The operation moves 626 General and 94 Agentic cards, for 720 cards in total. Physical AI contains no HOLD cards and its 182 locked mappings remain byte-equivalent apart from the release identifier.

This review path is orthogonal to semantic classification. For every moved card, the former L2 and L3 identifiers, bilingual labels, definitions, references, metrics, and migration metadata are preserved in `hold_semantic_path`. Accordingly, the public hierarchy has four canonical L2 display categories and 52 navigable L3 nodes: 50 semantic risk families plus two review-path nodes. The HOLD nodes must not be counted as new risk concepts or used as semantic prototypes in reliability analysis.

4 HOLD decision policy

4.1 Meaning and state transition

`HOLD` is the public rendering of `decision_required=true`. In v2.15.0 it is a review-path L2 under General and Agentic AI, not a semantic risk family. A held card has one navigable HOLD path and retains its prior semantic L3 candidate as explicit lineage metadata; neither path is human-approved ground truth. The state is set by a hard qualitative trigger or by a pending algorithmic-review trigger. It is cleared only by a recorded human decision that reconciles the label, mechanism-only definition, evidence, and destination definition; absence of a competing score alone is not sufficient.

4.2 Quantitative criteria

Quantitative measurements control automatic eligibility and review priority, not semantic truth. The following rules are applied or reported:

1. Stage 1 automatic acceptance requires $s_{sem} \geq 0.55$, $s_{comp} \geq 0.54$, both margins at least 0.035, at least two of three anchor votes, exact agreement of the rule, semantic, and composite winners, and two matching expert-agent approvals. Failure prevents automatic clearance.

2. Stage 2 uses the deterministic suitability score S_2 defined above; the lowest-ranked reserve and every hard hold retain decision metadata after forced operational assignment.

3. A constrained-EM move requires composite improvement of at least 0.020 over the source and keyword cosine of at least 0.015. Passing these thresholds permits a proposal but never clears `HOLD` automatically.

4. For v2.12.0 audit prioritization, a top-two cosine margin below 0.020 is high priority, a margin from 0.020 to below 0.050 is medium priority, and a margin of at least 0.050 is lower priority. Released-L3 rank outside the top three or per-card agreement below 80.0% under $\sigma = 0.05$ perturbation is an additional instability signal.

5. Strong geometric support is reported only when top-1 containment is at least 90.0%, top-2 containment at least 95.0%, the matched-size permutation test has $p < 0.001$, and perturbation agreement remains at least 97.0% through $\sigma = 0.05$. These are validation thresholds, not substitutes for the qualitative tests below.

4.3 Qualitative criteria

A card remains or becomes `HOLD` when any of the following substantive conditions applies:

- the label and mechanism definition describe different risks, or the cited source does not directly support the stated failure mode;

- the card is multi-mechanism, the destination L3 is overloaded, or two plausible L3 paths cannot be separated without a policy judgment;
- the risk exposes a taxonomy gap, so the present L3 is merely the closest operational path rather than a definitionally adequate family;
- an Agentic destination lacks an intrinsically Agentic execution loop, tool action, persistent memory, autonomous planning, or oversight mechanism;
- a non-authoritative card appears Physical but is outside the locked 182-card Physical set, or a proposed change would disturb a locked Physical path;
- independent reviewers reject the proposal, disagree on the destination, or do not reach exact high-confidence consensus; or
- the card was moved by constrained EM and has not yet received explicit human approval.

4.4 v2.12.0 HOLD baseline and v2.13.0 delta

The 689 v2.12.0 held cards are grouped by the stored primary reason in Table 4. Composite taxonomy-gap labels are counted once in the taxonomy-gap row.

Table 4: Primary reasons for the v2.12.0 HOLD queue.

Primary reason group	Cards
Taxonomy gap	280
Anthropomorphism mechanism not established	173
Overloaded L3 or low evidence fit	138
Expert review without consensus	28
Constrained-EM remap pending review	20
Physical-boundary conflict	16
Expert rejection	16
Expert disagreement	7
Multiple mechanisms	5
Other named exceptions	4
Low absolute fit	2
Total	689

Release v2.13.0 adds three review markers without changing any L3 path. Two cards require adjudication of a semantic near-duplicate scope boundary, and one additionally requires direct source-entailment review. The current queue therefore contains 692 cards (40.4%).

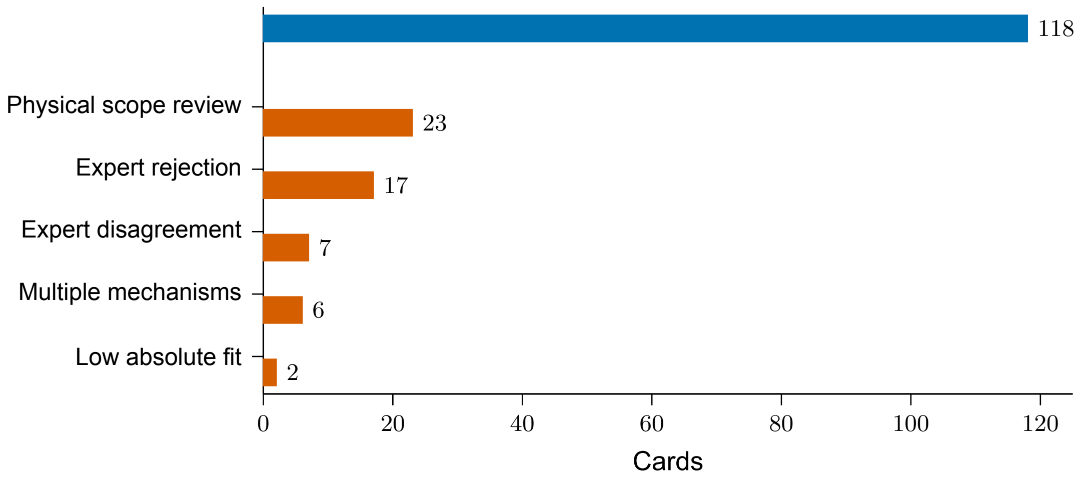

Figure 2: **Disposition of the 173 Stage 2 holds.** The 55 difficult cases (vermillion) are force-assigned and marked `HOLD`; 118 suitability-reserve cards (blue) are also force-assigned.

5 Results

5.1 Evidence composition

Papers account for 44.8% of the integrated AI evidence space, policy reports for 3.2%, and patents for 52.0%. Within the AI-risk evidence view, the corresponding shares are 60.7%, 1.1%, and 38.2%. These shares use one decimal because their integer component is at least one.

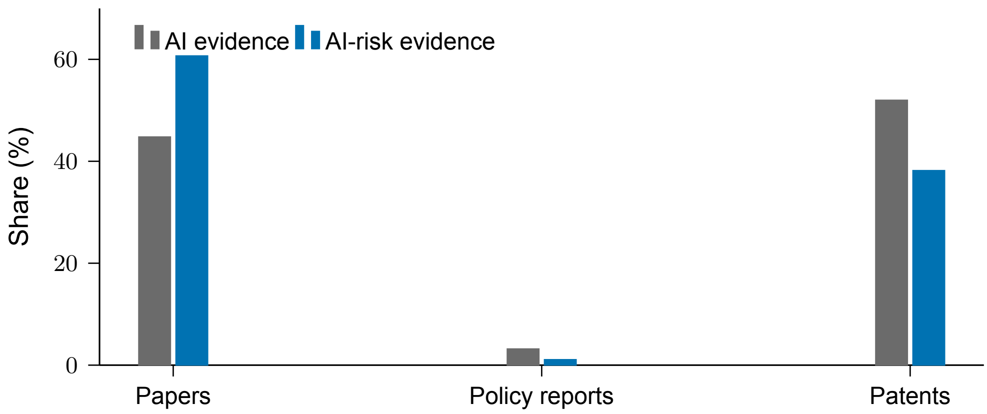

Figure 3: **Source-family composition.** Percentages compare document-type composition within the integrated AI evidence space and the AI-risk evidence view.

5.2 Final hierarchy coverage

The final policy assigns all 1,711 canonical cards (100.0%); 720 assigned cards (42.1%) carry the `HOLD` marker. Of all cards, 1,193 are General, 336 are Agentic AI, and 182 are Physical AI. Their shares are 69.7%, 19.6%, and 10.6%, respectively. All 50 semantic L3 nodes contain at least one non-HOLD card. The two HOLD review-path L3 nodes contain 626 General and 94 Agentic cards, respectively. Anthropomorphism is the largest L3 with 234 cards (13.7%); the five largest L3 nodes jointly contain 41.6%.

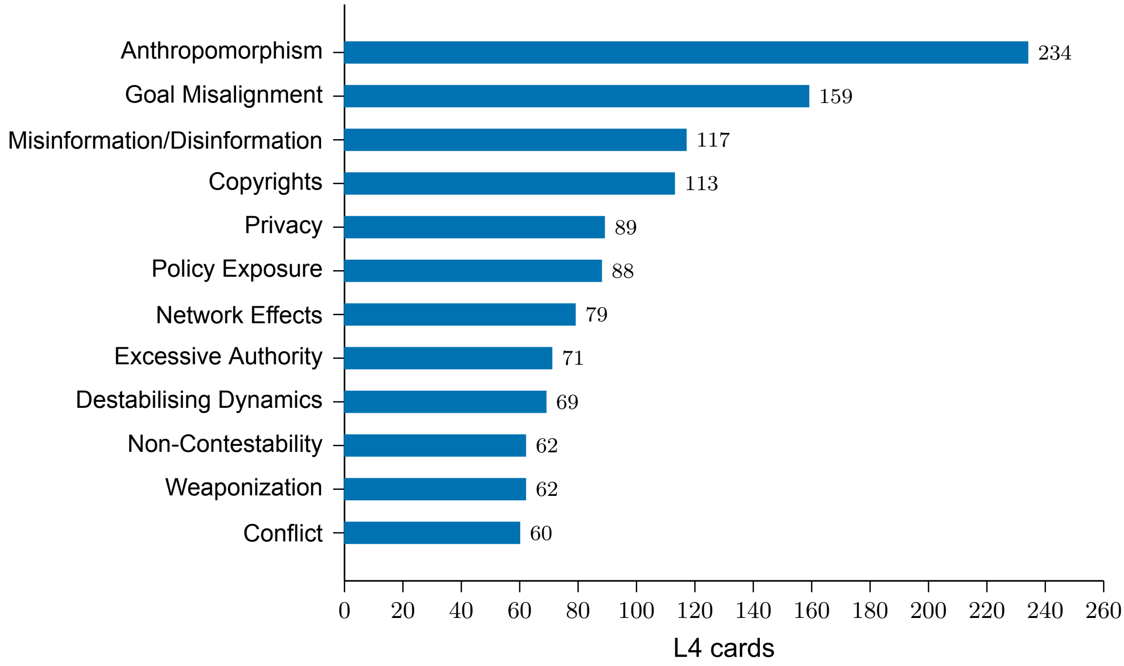

Figure 4: **Largest L3 nodes in the final policy view.** The figure shows the eleven largest L3 nodes. Counts include assigned cards marked `HOLD`.

6 Rounding and reporting policy

All counts are reported as integers. Percentages below 1 are rounded at the third decimal place and displayed to two decimal places; percentages with an integer component of at least 1 are rounded and displayed to one decimal place. For example, the L4 registry represents 0.04% of the integrated AI evidence count, the AI-risk view represents 2.1%, final L3 coverage is 100.0%, and the decision-required share is 42.1%. Exact floating-point ratios remain available in the machine-readable statistics artifact.

7 Assignment reliability validation

7.1 Reproducible procedure

Reliability is re-evaluated on all 1,711 v2.14.0 paths using dense, 1,024-dimensional BGE-M3 embeddings, seed 20260721, maximum sequence length 256, batch size 32, and L2 normalization. Card text concatenates bilingual L4 labels and definitions; L3 seed text concatenates bilingual L3 labels and definitions. The procedure reproduces four checks from the prior Technical Report. This rerun encodes the current v2.14.0 card text and all 50 active L3 seed definitions. It therefore incorporates both the v2.13.0 Memory wording revisions and the v2.14.0 forced migrations.

First, seed-initialized spherical EM alternates nearest-centroid assignment and normalized centroid recomputation:

$$z_i^{(t+1)} = \arg\max_k y_i^\top \mu_k^{(t)}, \qquad \mu_k^{(t+1)} = \frac{\sum_{i:z_i=k} y_i}{\left\|\sum_{i:z_i=k} y_i\right\|_2}. \tag{10}$$

Second, top-k containment is the proportion of cards whose published L3 is among the k nearest centroids calculated from the published partition. Third, the observed mean within-family cosine is compared with 5,000 label permutations that exactly preserve family sizes; the p-value uses the plus-one correction. Fourth, isotropic Gaussian embedding perturbation is repeated 200 times per σ, and agreement compares perturbed with unperturbed nearest-centroid assignments.

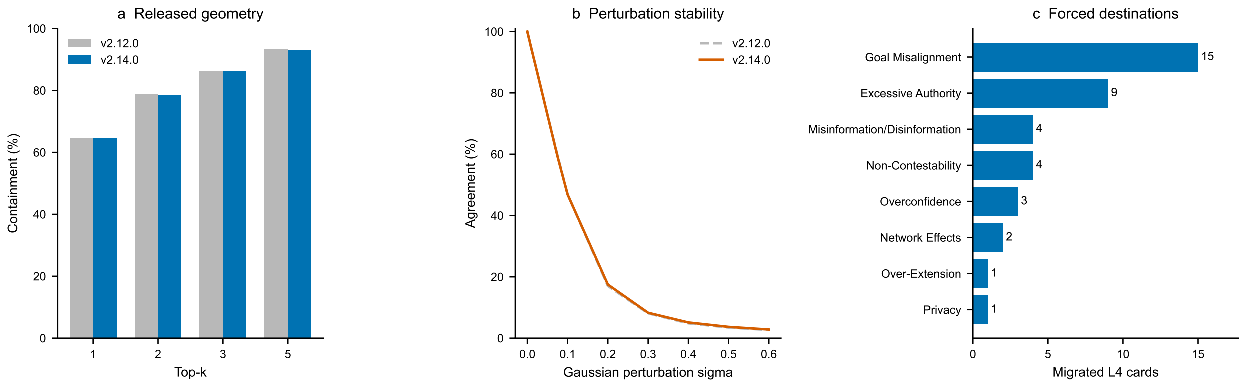

Figure 5: **Convergence and stability before HOLD separation (v2.14.0).** All 1,711 cards are assessed while the 720 HOLD cards remain embedded in their semantic L3 candidates. (a) Seed-initialized spherical EM reaches a fixed point after 15 iterations. (b) v2.14.0 top-k containment is compared with the v2.12.0 54-family baseline. (c) Perturbation stability remains materially similar after consolidation.

Table 5: Full-release reliability results before and after L3 consolidation.

Metric	v2.12.0	v2.14.0
L3 families	54	50
Cards assessed	1,711	1,711
Mean within-family cosine	0.770	0.768
Median assignment margin	0.012	0.012
Top-1 containment	64.6%	64.7%
Top-2 containment	78.7%	78.6%
Top-3 containment	86.1%	86.1%
Top-5 containment	93.3%	93.1%
Permutation p-value	0.0002	0.0002
Agreement at $\sigma = 0.01$	94.4%	94.5%
Agreement at $\sigma = 0.05$	72.8%	72.8%

The EM objective increases monotonically from 0.661 to 0.800 and reaches zero reassignments after 15 iterations. The converged unsupervised partition, however, agrees with only 21.8% of published paths (adjusted Rand index 0.120; normalized mutual information 0.402) and proposes 1,338 changes. Numerical convergence is therefore not evidence that the published semantics are correct. Published cohesion is 0.7684, above the matched-size null mean of 0.7382 with zero exceedances in 5,000 permutations (plus-one $p = 0.0002$). The taxonomy contains non-random semantic structure, but its global partition is not robust under the tested geometry.

7.2 Retired-family migration diagnostic

Table 6: Released-centroid support for the v2.14.0 forced migrations.

Subset	Cards	Mean cosine	Median margin	Top-1	Top-3
Migrated v2.14.0	39	0.746	-0.005	46.2%	71.8%
All other cards	1,672	0.769	0.013	65.1%	86.4%

The migrated subset is weaker than the unchanged population: 53.8% of its cards have negative assignment margins, and its top-1 containment is only 46.2%. This result supports the conservative policy decision to retain `HOLD` on every forced migration. It does not invalidate the operational destinations; it identifies them as priority items for human adjudication.

7.3 Decision and limitations

Sparse-family consolidation: OPERATIONAL PASS WITH REVIEW. Exactly 39 cards were migrated, no Physical path or L4 identity changed, all four retired families were archived, and every migrated card remains `HOLD`. The weak migrated-subset geometry prohibits interpreting the destinations as approved ground truth.

Full 50-family taxonomy: NOT YET DEMONSTRATED. The permutation test establishes non-random structure, but global containment, perturbation stability, and EM-to-published agreement do not satisfy strong reliability criteria. The weakest top-1 families are Destabilising Dynamics (36.2%), Goal Misalignment (38.2%), Network Effects (41.6%), Conflict (50.0%), Copyrights (53.1%), and Inconsistency (57.1%). Anthropomorphism remains large and heterogeneous at 234 cards and 65.0% top-1 containment.

Four limitations constrain interpretation. First, cross-category overlap in the evidence space means that source counts are analytical views, not mutually exclusive samples. Second, the transition from 82,971 records to 1,711 L4 cards changes the unit of analysis. Third, embedding geometry is a diagnostic, not a substitute for expert judgment or causal evidence. Fourth, small families can obtain perfect containment with limited statistical power.

7.4 Reliability after excluding the HOLD review paths, v2.15.0

The second analysis uses the identical encoder, seed, centroid procedure, 5,000 matched-size permutations, and 200 perturbation repeats, but excludes all 720 HOLD cards before constructing family centroids. It assesses 991 non-HOLD cards across the same 50 semantic L3 families; none of those families is empty. The two HOLD review nodes are navigation constructs and are not embedded or treated as risk-family prototypes.

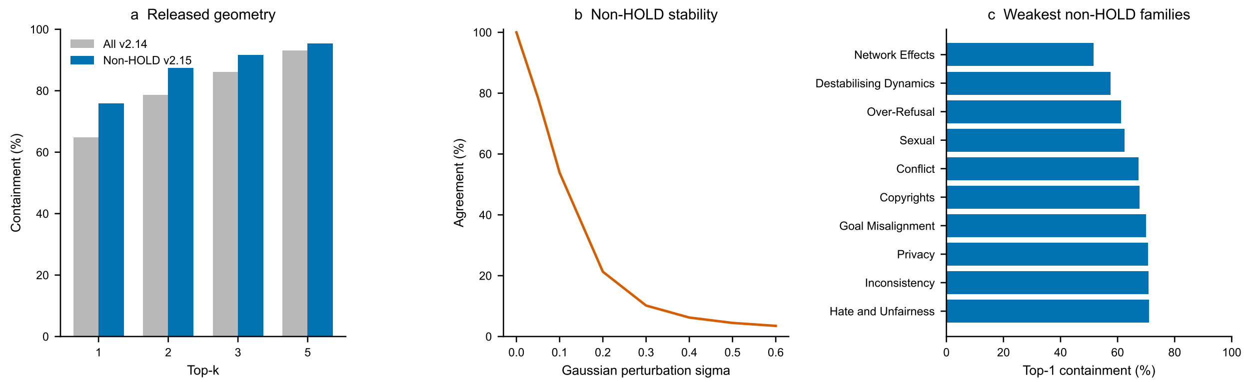

Figure 6: **Reliability after HOLD separation (v2.15.0).** (a) Top-k containment for the 991 non-HOLD cards is shown against the full v2.14.0 population. (b) Perturbation stability is calculated only on non-HOLD centroids and cards. (c) The ten weakest non-HOLD families identify the next review priorities.

Table 7: Reliability with HOLD embedded and after HOLD exclusion.

Metric	All v2.14.0	Non-HOLD v2.15.0
Cards assessed	1,711	991
Semantic L3 families	50	50
Mean within-family cosine	0.768	0.779
Median assignment margin	0.012	0.024
Negative-margin share	35.3%	24.2%
Top-1 containment	64.7%	75.8%
Top-2 containment	78.6%	87.3%
Top-3 containment	86.1%	91.5%
Top-5 containment	93.1%	95.4%
EM agreement with released paths	21.8%	31.7%
Agreement at $\sigma = 0.01$	94.5%	95.6%
Agreement at $\sigma = 0.05$	72.8%	78.4%
Permutation p-value	0.0002	0.0002

The conditional non-HOLD subset is more coherent: top-1 containment rises by 11.1 percentage points, the median margin approximately doubles, and the negative-margin share falls by 11.1 points. EM reaches a fixed point after 13 iterations and agrees with 31.7% of released paths (adjusted Rand index 0.187; normalized mutual information 0.501). Observed cohesion is 0.779 versus a matched-size null mean of 0.742, with zero exceedances and plus-one $p = 0.0002$.

These are not before–after accuracy estimates on the same population. The later analysis deliberately removes the review-priority cases that were known to have weaker fit, changes every centroid, and reduces the sample from 1,711 to 991. The improvement therefore validates HOLD as an effective triage separation signal; it does not prove that every non-HOLD assignment is correct or that moving a card into HOLD improves its semantic classification.

7.5 Independent specialist diagnosis of limited reliability

Two specialist agents independently reviewed the validation artefacts. One review concentrated on representation learning, spherical EM, and statistical estimands; the other examined ontology design, data provenance, and review policy. They were instructed to separate observations from causal inference and falsifiable hypotheses. Their agreement is an analytical review, not a substitute for the independent human gold-standard study recommended below.

The reviewers first cautioned that the three headline values estimate different properties and are not interchangeable accuracy measures. The 64.6% top-1 value is an *in-sample centroid-containment rate*: every card contributes to the released-family centroid against which it is evaluated. This construction is optimistic for small families and tautological for singletons, while its micro-average is heavily influenced by large families. The 72.9% agreement at $\sigma = 0.05$ measures sensitivity to a specified, uncalibrated correlated Gaussian perturbation of cosine scores; it is not an estimate of encoder reproducibility or assignment correctness. Finally, the 20.4% EM-to-release agreement compares an unconstrained spherical clustering objective with a normative expert taxonomy. It does not imply that 79.6% of released paths are erroneous.

Table 8: Two-reviewer diagnosis of the reliability estimands.

Diagnostic	Why the value can be limited	What the value does not establish
Top-1 containment, 64.6%	Overlapping L3 intensions, heterogeneous broad families, single-centroid misspecification, and in-sample family-size effects	Independent accuracy or a 35.4% error rate
$\sigma = 0.05$ agreement, 72.9%	The perturbation scale is about four times the median margin of 0.0123 and is not calibrated to observed encoder or text variation	Real-world reproducibility, semantic validity, or correctness probability
EM-to-release agreement, 20.4%	EM has no hierarchy, protected-path, family-size, causal-mechanism, or multi-label constraints and may drift from its seed meaning	That 79.6% of published paths should be moved
Permutation $p = 0.0002$	Large N yields a narrow null around a high cosine baseline; observed cohesion exceeds the null by 0.0314	Correctness, completeness, or uniqueness of the 54 families

The principal explanation is ontological rather than purely computational. The L3 layer mixes harm content, failure mechanisms, affected rights, governance processes, and interaction dynamics in one exclusive partition. Many L4 risks are therefore multi-axial: anthropomorphic trust can co-occur with privacy disclosure, manipulation, and non-contestability. Forced primary-only placement converts legitimate semantic multiplicity into narrow or negative margins. Boundary overlap is especially plausible between Goal Misalignment and Goal & Planning; Excessive Authority, Tool Calling, and Oversight & Control; and Network Effects, Destabilising Dynamics, Conflict, and Collusion. A single prototype per family cannot represent such multimodal intensions reliably.

The observed family distribution supports this diagnosis. Family sizes range from 1 to 234, with a median of 15.5; 19 families contain at most five cards, whereas 12 contain at least 50. Family size and top-1 containment are negatively correlated ($r = -0.70$), as are family size and median margin ($r = -0.52$). Anthropomorphism contains 234 cards, of which 229 remain `HOLD`; 173 lack an established anthropomorphic mechanism. Goal Misalignment contains 144 cards and has 38.2% top-1 containment with a negative median margin. Taxonomy gaps, an unestablished Anthropomorphism mechanism, and overloaded or low-fit destinations account for 591 of 689 HOLD cards (85.8%). These concentrations are consistent with large residual families absorbing risks for which a stable destination is absent or insufficiently defined.

Definition provenance is a second structural limitation. Among 1,529 non-Physical cards, 1,510 use the `source_mechanism_only_v2.7` definition form and 1,528 cite exactly one reference. Earlier remediation removed 1,129 instances of explicit upper-taxonomy scaffold prose, but truncating that prose does not by itself demonstrate that the retained atomic definition is entailed by the cited source. The embedding audit may consequently measure consistency of editorial abstractions as well as evidence-grounded risk identity. The stronger Physical result must also be interpreted separately: its 182 paths are authoritative locks with synchronized definitions, and 16 of its 24 families contain at most five cards. Its 89.0% top-1 result supports internal alignment but is not an unbiased comparison with non-Physical assignments.

The two reviews converge on four testable hypotheses. First, independently coded primary-plus-secondary mechanisms should be more common among low-margin cards if single-label pressure is causal. Second, leave-one-out or held-out centroids should reduce the apparent advantage of singleton and small families. Third, source-entailment rewriting should improve expert agreement and held-out margins if definition provenance is causal. Fourth, paraphrase, translation, truncation, and encoder-version experiments should yield an empirical score-drift distribution against which perturbation scales can be calibrated. Until those tests are completed, the current metrics support targeted review rather than wholesale algorithmic remapping.

Table 9: Reviewers' agreed validation priorities.

Priority	Intervention	Acceptance evidence
P0	Construct a stratified human gold set by L1, L3 size, HOLD, and boundary status; allow one primary and optional secondary L3	At least two domain experts per card, adjudication, agreement coefficient with uncertainty, and micro/macro accuracy with confidence intervals
P0	Rewrite L3 intensions as inclusion rules, exclusions, and contrastive examples; audit every sampled label–definition–reference triple for direct source entailment	At least 80% blinded agreement on boundary contrast sets; unsupported cards rewritten, split, or retired
P1	Review the 591 HOLD cards in the three dominant reason groups and resolve Anthropomorphism residuals without automatic cluster splitting	Documented migration matrix and repeated before/after reliability analysis
P1	Enforce Agentic-necessity predicates and evaluate primary plus secondary mechanisms	Generic model risks remain General; boundary-review agreement at least 0.70
P2	Replace resubstitution scoring with leave-one-out or held-out prototypes; report macro and micro estimates, confidence intervals, and empirical text perturbations	No self-inclusion; families with $n < 5$ marked insufficient; noise scale calibrated from observed variation

7.6 HOLD separation and protected Physical subset

The current HOLD queue is measurably weaker than the non-HOLD subset under the same released-centroid geometry. This supports the review marker as a triage instrument, although it does not prove that every individual marker is correct.

Table 10: v2.14.0 reliability by policy subset.

Subset	Cards	Mean cosine	Median margin	Top-1	Top-3
Physical locked	182	0.817	0.038	89.0%	97.3%
Non-Physical	1,529	0.763	0.009	61.8%	84.8%
HOLD	720	0.759	0.004	56.1%	82.6%
Non-HOLD	991	0.775	0.018	70.9%	88.6%
Forced migrations	39	0.746	-0.005	46.2%	71.8%

The HOLD top-1 rate is 14.8 percentage points below the non-HOLD rate, and its median margin is less than one quarter of the non-HOLD margin. Negative margins occur for 43.9% of HOLD cards versus 29.1% of non-HOLD cards. The protected Physical subset is substantially stronger, with 89.0% top-1 and 96.7% top-2 containment, but remains locked because authority comes from the Physical AI taxonomy rather than from this geometric diagnostic.

Relative to v2.12.0, v2.14.0 raises global top-1 containment by 0.1 percentage points, while cohesion decreases by 0.0018, top-2 decreases by 0.1 points, and $\sigma = 0.05$ agreement is unchanged at the displayed precision. These differences are negligible at global scale. The correct conclusion is therefore **SHARE WITH CAVEATS**: the taxonomy has statistically non-random semantic structure and a useful HOLD separation, but strong global assignment reliability is not demonstrated.

8 Review priorities

Review should begin with the 39 forced migrations, then the weakest global families and low-margin cards within Anthropomorphism. The 720 assigned cards marked `HOLD` should be adjudicated using agreement among the risk label, mechanism-only definition, cited evidence, and candidate L3 definition. Each approved batch should be followed by the identical reliability analysis. A low score alone must not trigger automatic movement; human review determines whether to retain the path, expand an L3 definition, or propose a genuinely necessary new L3.

This document is the English journal-style edition of *RAI Risk Taxonomy Technical Report 2.0*.